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Question 1

Our analysis uses a dataset of credit card information gathered from 3000 individuals. This dataset includes demographic variables such as gender, education level, age, and marriage status, as well as financial variables like recent bill amounts, amount of credit limit, repayment status over the past months, and payment amounts made on previous bills. Repayment status variables indicate whether a payment was made on time or how many months it was delayed. The goal of this analysis is to build a model that identifies whether a customer is likely to default on their credit card payment in the following month.

Question 2

To select the best classification model for predicting credit card default, four different models were compared: a logistic regression model, a k-nearest neighbors (KNN) model, a decision tree with a maximum depth of 3, and a decision tree with a maximum depth of 5. The dataset was split into features and the target variable (“default payment next month”), and each model was evaluated using 8-fold cross-validation to ensure that performance was measured consistently across different subsets of the data. The primary evaluation metric used was average cross-validation accuracy, allowing for a fair comparison of how well each model generalizes to unseen data.

Once every model finished running, their average accuracy scores got lined up side by side - whichever had the top cross-validation score became the preferred pick. Relying on several data splits instead of just one random division keeps the choice steady over shifting samples. Interpretability entered the picture too, especially for decision trees; should one rise to the top, its feature importance values came under review. Financial factors driving default predictions then surfaced clearly through that lens. Accuracy stayed central, yes - but clarity mattered whenever it could fit without compromise.

Question 3

The decision tree model with a maximum depth of 3 was the model that we felt like was the best. Decision trees divide the feature space by splitting on the feature and threshold that best separates defaulters from non-defaulters at each node. The tree can only have three levels of splits because the maximum depth is three. This keeps the model simple instead of becoming too complicated and reduces the risk of overfitting as a result. The model finds that the most predictive features were the variables PAY_0, PAY_2, and PAY_AMT3. The model had a pretty high accuracy of 82.05%, meaning that it is accurate about 4 out of 5 times. These results show that the decision tree with the depth of 3 is the best model to choose for managing credit risk out of the ones listed.

Appendix 1

```
# Importing necessary libraries
import numpy as np
import pandas as pd
from sklearn.linear_model import LogisticRegression
import statsmodels.api as sm
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import confusion_matrix, accuracy_score, precision_score, recall_score, f1_score
from sklearn.neighbors import KNeighborsClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import cross_val_score
```

```

# Read data
data = pd.read_csv("https://webpages.charlotte.edu/mschuck1/classes/DTSC2301/Data/credit_card_defaults.csv", na_values=['NA'])

# Select features
feature_names = ['ID', 'LIMIT_BAL', 'SEX', 'EDUCATION', 'MARRIAGE', 'AGE', 'PAY_0', 'PAY_2', 'PAY_3', 'PAY_4', 'PAY_5', 'PAY_6', 'BILL_AMT1', 'BILL_AMT2',
'BILL_AMT3', 'BILL_AMT4', 'BILL_AMT5', 'BILL_AMT6', 'PAY_AMT1', 'PAY_AMT2', 'PAY_AMT3', 'PAY_AMT4', 'PAY_AMT5', 'PAY_AMT6']
X = data[feature_names]
y = data['default payment next month']

# Models
models = {
    'Logistic Regression': LogisticRegression(),
    'k-NN': KNeighborsClassifier(),
    'Decision Tree (depth 3)': DecisionTreeClassifier(max_depth=3),
    'Decision Tree (depth 5)': DecisionTreeClassifier(max_depth=5)
}

# 8-fold cross-validation
cv_scores = {}
kf = KFold(n_splits=8, shuffle=True, random_state=42)
for name, model in models.items():
    scores = cross_val_score(model, X, y, cv=kf)
    cv_scores[name] = scores.mean()
    print(f"{name} average CV accuracy: {scores.mean():.4f}")

# Best model interpretation
best_model_name = max(cv_scores, key=cv_scores.get)
print(f"\nBest model: {best_model_name} with accuracy {cv_scores[best_model_name]:.4f}")

# For decision trees, feature importances
if 'Decision Tree' in best_model_name:
    model = models[best_model_name]
    model.fit(X, y)
    importances = model.feature_importances_
    sorted_indices = importances.argsort()[::-1]
    print("Most useful features:")
    for i in sorted_indices[:3]: # Top 3
        print(f"{feature_names[i]}: {importances[i]:.4f}")

```

Logistic Regression average CV accuracy: 0.7788

k-NN average CV accuracy: 0.7493

Decision Tree (depth 3) average CV accuracy: 0.8205

Decision Tree (depth 5) average CV accuracy: 0.8204

Best model: Decision Tree (depth 3) with accuracy 0.8205

Most useful features:

PAY_0: 0.7672

PAY_2: 0.1515

PAY_AMT3: 0.0489